

ChristosHadjinikolis

Engineering Manager | Developer Infrastructure & Distributed Systems

Ph.D. Computer Science | UCL Research Associate | AI Standards Expert @ ISO/CEN-CENELEC JTC'21

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mediums



people

coaching

feedback

hiring loops

retention/progression

distributed collaboration

developer
tools

PyPI libraries

JetBrains plugin

runtime replay

source navigation

AI-assisted review

platforms

distributed systems

event pipelines

high-volume streaming

cross-team dependency

quality

testing discipline

automation

observability

incident response

DORA/Jira signals

technical

Java/Kotlin

Python/SQL

Kafka/Flink

Airflow/AWS/GCP

logic

computational logic

formal semantics

Semantic Web

graph analytics

process mining

about me

Engineering Manager for developer infrastructure, distributed systems, and production platforms with **16 years** across backend engineering, data infrastructure, consulting, AI research, and teaching. I build engineering systems that help other engineers ship software safely, reliably, and at scale. At Vortexa, own a major **live streaming backend estate**, lead a **10-person cross-functional team**, and drove Data Production Team growth from **4 to 30+ people** by shaping hiring, system-design culture, platform choices, and operating discipline. My public work on **PyPI libraries**, a **JetBrains IDE plugin**, runtime replay, and AI-assisted review reflects the same bias: make complex software easier to understand, test, operate, and improve.

work-experience

Vortexa Ltd, London, UK 12/2020–present

Engineering Manager | ML Systems Lead

Own the **engineering strategy and delivery** for Vortexa's **live streaming intelligence estate**, a major client-facing backend area powering product and commercial workflows, filtering roughly **6M vessel-position records/hour** into production intelligence for **13.5K monitored vessels**. Lead **live-processing strategy** with a **10-person cross-functional team** spanning **6 MLE/DE/DS, 2 Product, and 2 SMEs**. The estate behaves like an internal platform: downstream teams and product surfaces depend on its correctness, freshness, reliability, and operational clarity.

- **People Leadership:** Directly manage the MLE/DE/DS core and lead a **10-person cross-functional delivery team**; hiring manager for **6 roles**, interviewing **31+ candidates** in that capacity and **60+ overall**; retained the full current team, with every member recently promoted.
- **Organisation Scaling:** Drove Data Production Team growth from **4 to 30+** through hiring loops, system-design evaluations, onboarding discipline, mentoring, reduced key-person risk, and clearer ownership boundaries.
- **Developer Infrastructure:** Improve engineering feedback loops through Flink/Airflow archetypes, dev containers, Backstage adoption, local E2E/integration tests, tests-as-docs, and ADR-backed architecture decisions; cut a **3h DAG feedback loop to under 5 min**.
- **Distributed Platform Strategy:** Enabled independent scaling of critical live pipelines by leading the two-year **Kafka Streams-to-Flink migration**; used Flink's dataflow model, independent state/checkpointing, richer operational UI, and framework-driven design to reduce maintenance effort.
- **Reliability & Operational Health:** Protect customer-facing reliability with monitoring, runbooks, rollback/fallback plans, Jira alert workflows, and shared on-call ownership while keeping **MTTR under 30 min**.
- **Telemetry & Delivery Quality:** Use DORA/Jira signals, accuracy dashboards, alert workflows, and service monitoring to make delivery and operational quality visible; latest view: **14.65 deploys/week** with **0.52% change failure**.
- **Architecture & Cross-Team Clarity:** Instigated a weekly company architecture forum; partner with Product, analysts, SMEs, and engineering stakeholders to turn ambiguous product/interface problems into measurable workstreams and clear dependency boundaries.

DataReply, London, UK 04/2016–12/2020

Senior Consultant | Data Scientist / ML Engineer

Joined Data Reply's London spin-off as its first consultant and played a formative role in growing it from **three practitioners and two managers to 30+ consultants**. Progressed to Senior Consultant; led client meetings, scoped goals, placed consultants, interviewed/mentored hires, and delivered data-platform and ML engineering work across UBS, Vodafone, CNHi and other enterprise clients. Ran the London Flink Meetup for five years.

- **Client Delivery Leadership:** Drove agendas with client stakeholders and product teams, agreed goals, facilitated technical delivery, and supported hiring/consultant placement across banking, telecoms, manufacturing, publishing, and gaming accounts.
- **Vodafone Group:** Led technical delivery across multiple cloud/data workstreams; worked on Infinity, a GCP/**Kubeflow** DS platform, shaping CI/CD, workflow automation, ML lifecycle support, and feature-engineering migration practices.
- **CNHi:** Lead Data Scientist/Scrum Master on predictive maintenance; managed/guided a DS team using **PySpark** over live vehicle sensory data with product-facing backlog discipline.
- **UBS:** Grew from Data Scientist into ML Engineer through XP/pairing while delivering graph/process-mining analytics, observability work, and real-time pipelines with **Kafka**, **Elasticsearch**, and Python.

Roles in UK Institutions 2010–2016

Coding Teacher @DGC | Teaching Assistant @KCL/UCL

Built the communication and presentation discipline I now rely on as a leader by teaching Java/Python and core computer science at David Game College from 09/2012 to 04/2016 while completing doctoral research; Teaching Assistant at KCL and UCL.

selected developer tooling

PyPI	skeleton-replay Python developer tool that turns script/pytest runs into traces, architecture snapshots, workflow evidence, and replayable reports for review, debugging, onboarding, and LLM-assisted understanding.	Runtime architecture evidence
JetBrains	Skeleton Replay plugin Free PyCharm/IntelliJ plugin bringing Skeleton reports, workflow evidence, and source navigation into the IDE; reflects a bias for developer experience, evidence, and shared understanding.	IDE workflow
PyPI	dynamicio Free public tooling for making I/O seams explicit, validating schema/data expectations, and switching local/dev/prod datasets so pipeline logic can be tested with faster feedback.	ML/data I/O contracts

education

2010–2014	Ph.D. in Computer Science <i>Thesis: Persuasion Dialogues & Opponent Modelling.</i> Developed logical inference methods for large Knowledge Graphs , combining symbolic reasoning , graph analytics , and Bayesian techniques (e.g., MCMC). Research contributed to scalable AI systems grounded in formal semantics . Received Best Poster Award at IJCAI 2013 (413 submissions). Awarded EPSRC PhD Scholarship , Outstanding TA Award (student-nominated), and earned the Graduate Certificate in Academic Practice . Access Publications on Google Scholar	King's College London, UK
2004–2010	Diploma (BEng) in Computer Engineering Five-year polytechnic degree in Computer Science and Engineering with a strong focus on mathematics (10 courses), covering theoretical aspects, networking, and telecommunications. <i>Majored in Artificial Intelligence.</i>	University of Thessaly, GR
2002–2004	Military Officer: Second Lieutenant Completed national service as a commissioned Second Lieutenant.	National Guard, CY

committees/affiliations

since 10/2024	University College London, Dept. of Information Studies Research/teach practical AI standardisation: AI Act, auditability, model/data versioning, explainability, and safe adoption.	Associate Researcher
since 01/2021	ISO - International Organisation for Standardisation Member of JTC'21 Working Group 3, developing AI standards aligned with EU policy and international norms.	Committee Expert Member for JTC'21

talks/interviews

2023	Bill Raymonds, Agile in Action Agile Data Science: Sailing Through Data Science: The Agile Journey at Vortexa , Details: YouTube	Podcast Interview
2022	Open Data Science Conference dynamicio , a published PyPI library for abstracting I/O in ML/data workflows, Details: here	Industry Conference Talk
2020	iunera: Big Data Innovators The Agile approach in data science explained by an ML expert, Details: here	Interview Blog Post
2020	Big Data Warsaw Monitoring & Analysing Communication and Trade Events as Graphs, Details: here	Industry Conference Talk
2018	Connected Data London AI tribes and interconnections: what's graph got to do with it?, Details: here	Panelist @ Industry Conference
2018	Minds Mastering Machines A talk on doing Data Science the Agile Way, Details: here	Industry Conference Talk

certificates & awards

07/2020	AWS Certified Machine Learning - Specialty Validation Number: D0R0WC7CN1VQ13SV	AWS
01/2019	Process Mining License number: AFU2VTP6273V	Coursera Course Certificates
10/2017	Google Certified Professional - Data Engineer Credential ID: 000000400	Google Cloud Certified
07/2017	Advanced Elasticsearch: Data modelling Enrolment Id: 33442	Elastic
06/2016	Graph Analytics for Big Data License number: NZADYAQ7N8FW	Coursera Course Certificates
05/2016	Apache Spark Developer Certification License number: 1.x-0548	O'Reilly Media
04/2016	Neo4j Certified Professional License number: here	Neo Technology, Creators of Neo4j